Reader: Sure, in the first example the ninth term must be 256, while in the second example it must be 21.

Author: Correct. It means that in all the examples there is a certain law, which makes it possible to write down the ninth, tenth, and other terms of the sequences. Note, though, that if there is a finite number of terms in a sequence, one may fail to discover the law which governs the infinite sequence.

Reader: Yes, but in our case these laws are easily recognizable. In example (1) we have the terms of an infinite geometric progression with common ratio 2. In example (2) we notice a sequence of odd numbers starting from 5. In example (3) we recognize a sequence of squares of natural numbers.

Sam: Q) How exactly are we easily recognizing the law? From where is the intuition or understanding coming from? Can we properly define the term "certain law"? Given any infinite sequence, can we design an algorithm to check whether it contains "a certain law" or not? If yes, can we find it out and enumerate all the numbers in the sequence? Is it necessary for the law to be unique?

Author: Now let us look at the situation more rigorously. Let us enumerate all the terms of the sequence in sequential order, i.e. $1, 2, 3, \ldots, n, \ldots$ There is a certain law (a rule) by which each of these natural numbers is assigned to a certain number (the corresponding term of the sequence). In example (1) this arrangement is as follows:

1	2	4	8	16	32	... 2^{n-1}	(terms of the sequence)
↑	↑	↑	↑	↑	↑	↑	
1	2	3	4	5	6	... n	(position numbers of the terms)

In order to describe a sequence it is sufficient to indicate the term of the sequence corresponding to the number n, i.e. to write down the term of the sequence occupying the nth position. Thus, we can formulate the following definition of a sequence.

Definition: We say that there is an infinite numerical sequence if every natural number (position number) is unambiguously placed in correspondence with a definite number (term of the sequence) by a specific rule.